Where Does the Prompt Go? Probing Prompt Adherence in Diffusion Models

Problem Statement

You give an image generator a very specific prompt:

> **“A small red cube and a large blue sphere on a wooden table. The cube is to the left of the sphere.”**

The model may produce a beautiful, realistic image but on closer inspection:

- the cube is **blue** and the sphere is red,
- the cube and sphere have the **wrong relative sizes**,
- the objects appear on the **wrong sides**,
- or one of the objects is missing entirely.

This is a common problem with text-to-image generation: **an image can look convincing while still failing to follow important parts of the prompt.**

The interesting question is not simply *“Can we make the image look better?”*, but:

> **What happens inside the diffusion model when it succeeds or fails to follow a prompt?**

In this project, students will investigate **prompt adherence** in text-to-image diffusion models, focusing on how inference-time mechanisms such as **Classifier-Free Guidance (CFG)** affect different kinds of prompt constraints.

The objective of this problem is to investigate **how inference-time mechanisms influence prompt adherence**, and to develop an experimentally grounded understanding of when and why these mechanisms succeed or fail.

Rather than simply comparing parameter values, participants are expected to conduct a **hypothesis-driven, mechanistic study** of the generation process.

Research Direction

Participants should formulate a **specific, testable hypothesis** about how **classifier-free guidance (CFG)** and at least **one other inference-time parameter or mechanism** affect different forms of prompt adherence.

Possible dimensions of prompt adherence include:

- **Attribute binding:** Does the model correctly associate attributes with the intended object?
- **Object counting:** Does the generated image contain the requested number of objects?
- **Spatial relationships:** Are objects positioned according to the prompt?
- **Object presence:** Are all requested entities generated?
- **Compositional complexity:** How does adherence change as the number and interaction of constraints increase?

The hypothesis should make a **specific prediction**, rather than simply stating that changing a parameter will affect performance.

Participants are expected to formulate and justify their own hypothesis based on their understanding of diffusion models and prior literature.

Mechanistic Analysis

In addition to evaluating the final generated images, participants should investigate **what happens inside the model**.

In particular, the study should examine **cross-attention maps** to investigate how textual tokens influence the generated image during the denoising process.

Participants may investigate questions such as:

- How does attention to relevant tokens change across denoising timesteps?
- Do correctly generated attributes correspond to stronger or more localized attention?
- Does CFG alter the cross-attention behaviour?
- Are attention patterns different for successful and failed generations?
- Can attention-based observations help explain failures in attribute binding, counting, or spatial relationships?

The goal is to connect:

inference-time parameter → internal model behaviour → observable generation quality

rather than treating the diffusion model purely as a black box.

Experimental Requirements

1. Hypothesis formulation

Before running the main experiments, participants should:

- formulate a clear hypothesis;
- explain the mechanism they expect to produce the predicted behavior;
- identify what observations/results would **support or contradict** the hypothesis.

---

2. Controlled prompt benchmark

Participants should construct a controlled set of prompts representing different types and levels of constraints.

For example:

Category	Example
Object presence	"A dog and a bicycle"
Attribute binding	"A small red cube and a large blue sphere"
Counting	"Three yellow apples on a table"
Spatial relation	"A red circle to the left of a blue square"
Multiple constraints	"Three small red balls arranged in a triangle"

The prompt set **must be justified**.

Participants should explain:

- Why these particular prompt categories were selected.
- What aspect of prompt adherence each category tests.
- How prompt complexity is defined and controlled.
- Why the chosen prompts provide a meaningful test of the hypothesis.

Where possible, prompts should be generated systematically rather than selected purely by subjective judgement.

---

3. Controlled parameter study

Participants should vary **CFG and at least one additional inference-time parameter or mechanism** while keeping other experimental conditions controlled.

Possible choices include:

- denoising/sampling steps,
- noise or scheduler-related parameters,
- conditioning strength,
- sampling strategy,
- Cross attention manipulation
- Attention re-weighting
- Latent-space interventions

  The choices of the parameters should be **motivated by the hypothesis and the underlying diffusion mechanism**, rather than chosen simply because they are easy to change.

---

4. Multiple seeds and statistical analysis

Each experimental condition must be evaluated over **multiple random seeds** rather than a single generated image.

For example:

Prompt * CFG * Parameter * Seed

This is important because diffusion generation is stochastic, and a result observed for one seed may simply be an artifact of that particular generation.

Participants should report:

- mean performance,
- variance / standard deviation or confidence intervals,
- effect sizes where appropriate,
- and an appropriate statistical test for the comparisons being made.

The final conclusion should therefore be based on **statistical evidence across generations**, rather than individual examples.

Evaluation

Participants should define quantitative metrics for each type of prompt constraint.

For example:

Prompt adherence

- Object presence accuracy
- Counting accuracy
- Attribute-binding accuracy
- Spatial-relation accuracy
- Overall constraint satisfaction

Generation quality

- Image-text similarity
- Image quality/aesthetic metrics where appropriate

Diversity

- Variation across seeds

Mechanistic analysis

- Cross-attention distribution over relevant tokens
- Attention localization
- Changes in attention across denoising timesteps
- Relationship between attention patterns and successful/failed generations

The exact metrics should be justified rather than arbitrarily chosen.

Expected Outcome

The objective is **not necessarily to obtain a particular result** or to demonstrate that a higher CFG is always better.

Participants should determine:

1. Whether their hypothesis is supported or rejected by the experimental evidence.
2. How CFG and the second investigated mechanism affect different types of prompt constraints.
3. Whether the observed changes in prompt adherence can be related to changes in the model's cross-attention behavior.
4. Whether a simple inference-time intervention can improve prompt adherence without retraining the underlying diffusion model.

A successful project should produce a **quantitative characterization of the observed behavior**, supported by statistical analysis and mechanistic evidence.

Suggested tools

- **Hugging Face Diffusers** — model loading, generation and inference control
- **PyTorch** — experimentation and analysis
- **Python / Jupyter or Colab** — experimentation and visualization
- Tools for extracting and visualizing **cross-attention maps**

Resources (yet to add)
Provide starter code or a baseline notebook, a simple Stable Diffusion script with CFG variation and attention map extraction.

Starter baseline

Notebook containing-

- Loading a pretrained Stable Diffusion model using Hugging Face Diffusers
- Text-to-image generation
- Fixed/random seed control
- CFG variation
- Batch generation across prompts and seeds
- Saving generated images and experiment metadata
- Basic prompt-adherence evaluation
- A starter implementation for extracting and visualizing cross-attention maps

This means the challenge becomes about research and experimentation, rather than spending the first two weeks figuring out how to hook into a UNet.

Suggested literature/code for cross-attention

- Prompt-to-Prompt: Image Editing with Cross-Attention Control - particularly useful for understanding how cross-attention is accessed and manipulated during diffusion. The official implementation includes attention-control code and notebooks.

  https://github.com/google/prompt-to-prompt/

- Attend-and-Excite: Attention-Based Semantic Guidance for Text-to-Image Diffusion Models - particularly relevant because it directly studies failures such as missing subjects and incorrect attribute binding, and provides notebooks for cross-attention visualization.

  https://github.com/yuval-alaluf/Attend-and-Excite

- DAAM: Diffusion Attentive Attribution Maps - useful as an additional reference for extracting and interpreting token-level cross-attention in Stable Diffusion, with an available implementation and Colab notebook.

  https://github.com/castorini/daam